

Validity Analysis: MMLU

Target context: North India Competitive Exam Aspirants (Hindi-Medium)

Overall risk: **HIGH**

Deployment Context

Use case: Deployment scenario: A Hindi-speaking student in India is testing his/her preparation for competitive job-related examinations, with an AI system providing feedback on their responses. The goal is to evaluate the applicability of the benchmark and the quality of the LLM's feedback.

Domain: Educational assessment

Setting: Mobile application / enterprise software

Note: The deployment hypothesis should be tested using Hindi-language sentences from the benchmark, as I am familiar with the language and can provide evaluation in a subsequent stage.

Target population: A graduate student in North India interacting in the Hindi language. The student has completed their education primarily in Hindi, with limited exposure to English.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology	1	Serious concern	high
Input Content	1	Serious concern	high
Input Form	1	Serious concern	high
Output Ontology	1	Serious concern	high
Output Content	1	Serious concern	high
Output Form	2	Significant gaps	high
Average	1.2		

= highest concern = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MMLU is a near-total mismatch for the North India Hindi-medium competitive exam deployment across five of six validity dimensions. Subject coverage is anchored in US/Western academic curriculum with zero Indian Polity, Indian History, Indian

Geography, Current Affairs, or Hindi-language content (the high-priority domains identified in elicitation). Individual items are culturally embedded in US institutional, legal, demographic, and moral contexts; where India does appear, it is filtered through colonial or Western-pollster framing incompatible with the NCERT/Bipin Chandra-aligned exam-prep tradition. The benchmark is entirely English with no Devanagari content, conflicting with the deployment's Hindi-dominant input requirement. Output is label-only with no rationale evaluation, while the deployment requires substantive Hindi explanatory feedback. Annotator pool is undocumented but US-anchored; moral_scenarios labels are explicitly framed in US 2020 norms. Output form is the only dimension with a partial structural match (MCQ format), but even that is undermined by calibration failures and absence of generative-quality measurement. Indian-grounded alternatives (MILU, ParamBench, IndicEval, IndicMMLU-Pro) exist and should be primary; MMLU at best provides weak proxy signal on STEM/reasoning subdomains.

Practical Guidance

What This Benchmark Measures

MMLU measures broad English-language multitask classification accuracy on a US/Western academic curriculum, with strongest signal on culturally-neutral STEM (mathematics, physics, computer science) and weakest on humanities, law, and morality. For the NI-CEA deployment, MMLU can provide only weak proxy evidence for the Mathematics & Reasoning domain (one of several high-priority areas) and provides no signal at all on Indian Polity, Indian History, Indian Geography, Current Affairs, Hindi proficiency, or Hindi explanatory feedback quality.

Construct Depth

MMLU probes breadth across 57 subjects but at modest depth (5-shot prompts, label-only scoring [Q34, Q44]). For the deployment's core constructs — India-grounded factual knowledge, Hindi-medium comprehension, and Hindi pedagogical explanation — MMLU's depth is effectively zero. The English-vs-Hindi performance gap documented in IndicEval [WEB-16] confirms that MMLU's English-medium scores do not predict Hindi-medium ability even on overlapping content.

What Else You Need

Substantial supplementation is required. (1) For Input Ontology / Input Content, use MILU [WEB-13, WEB-14] and ParamBench [WEB-15] as primary benchmarks; they cover UPSC/SSC-relevant Indian curriculum content in Hindi. (2) For Input Form, use IndicEval [WEB-16] or IndicMMLU-Pro [WEB-18] to measure Hindi-medium degradation. (3) For Output Ontology / Output Form (Hindi explanatory feedback), commission a custom rubric — no existing benchmark covers this gap [WEB-21]. (4) For Output Content, conduct Indian-SME annotation review on any MMLU items used. (5) For Current Affairs, build a continuously-updated PIB/Yojana-sourced evaluation set, since static benchmarks cannot cover this domain.

ID	Evidence
Q34	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q44	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
WEB-13	MILU derives labels from official Indian competitive exams — implicit Indian-stakeholder validity that MMLU lacks (https://aclanthology.org/2025.naacl-long.507/)
WEB-14	https://huggingface.co/datasets/ai4bharat/MILU
WEB-15	ParamBench uses two-tier annotation team with Indian-context expert review — contrast model for what MMLU lacks (https://arxiv.org/abs/2508.16185)
WEB-16	IndicEval documents marked accuracy drops in Hindi vs. English (LLaMA 39.53% UPSC Hindi Zero-Shot) (https://arxiv.org/abs/2602.16467)
WEB-18	https://arxiv.org/abs/2501.15747
WEB-21	No Hindi exam-prep explanation-quality rubric exists (https://arxiv.org/abs/2508.19831)